Continental-Scale Streamflow Simulation Using Kriging in the NGIAB–NRDS NextGen Ecosystem

Sonali Vyas[1], Kunal Sarna[1], Suma Battula[2], Jonathan Frame[2]

[1]Computer Science · [2]Geological Sciences · The University of Alabama

Background

• Process-based models (NextGen, NWM) can only be evaluated where USGS gauges exist, yet >95% of the CONUS river network is ungauged.

• Need: a data-driven, observation-based framework with spatially complete streamflow estimates AND quantified uncertainty independent of model structure.

• Recent CIROH work ("Developing & Benchmarking DA Methods on a Standardized Testbed") shows ordinary kriging between USGS gauges is scalable AND accurate.

• Serves as a "pseudo-observation" for analysis, historical reconstruction, and data assimilation.

Hourly Kriging Method

• Daily-scale precedent: Farmer et al. 2019, National Hydrologic Model. Extended HERE to HOURLY.

• Input: instantaneous-value (IV) discharge from ~7,300 active USGS gauges (CONUS).

• IV → hourly UTC means with DST-aware tz_cd conversion (handles spring-forward + fall-back).

• Ordinary kriging with spherical variogram model:

nugget = 1 sill = 10 range = 100 km

• 200 × 200 CONUS grid (~50k pixels); mm/hr units.

• Outputs per hour: gridded interpolated runoff + kriging error variance.

• Sampled at every NextGen v2.2 catchment centroid → 831,777 catchments.

NRDS Integration (Operational)

• Hourly kriging runs as a first-class NRDS datastream (CIROH-UA).

• Nightly 00:30 ET → updates previous 24 hourly estimates per day.

• Pipeline: NWIS bulk fetch → 24-hr parallel kriging → per-catchment extraction → S3 publication.

• Per day: 24 NCs + variogram CSVs + plots/GIF + a single BROTLI-compressed Parquet (~13 MB; 831,777 × 24 rows).

• Public bucket: s3://ciroh-community-ngen-datastream/outputs/qkrig/qkrig.{YYYYMMDD}/

Application

• NWM calibration / evaluation

• Data assimilation using kriged as pseudo-observations for ungauged catchments (CFE EnKF)

• T-route initialization in ungauged basins

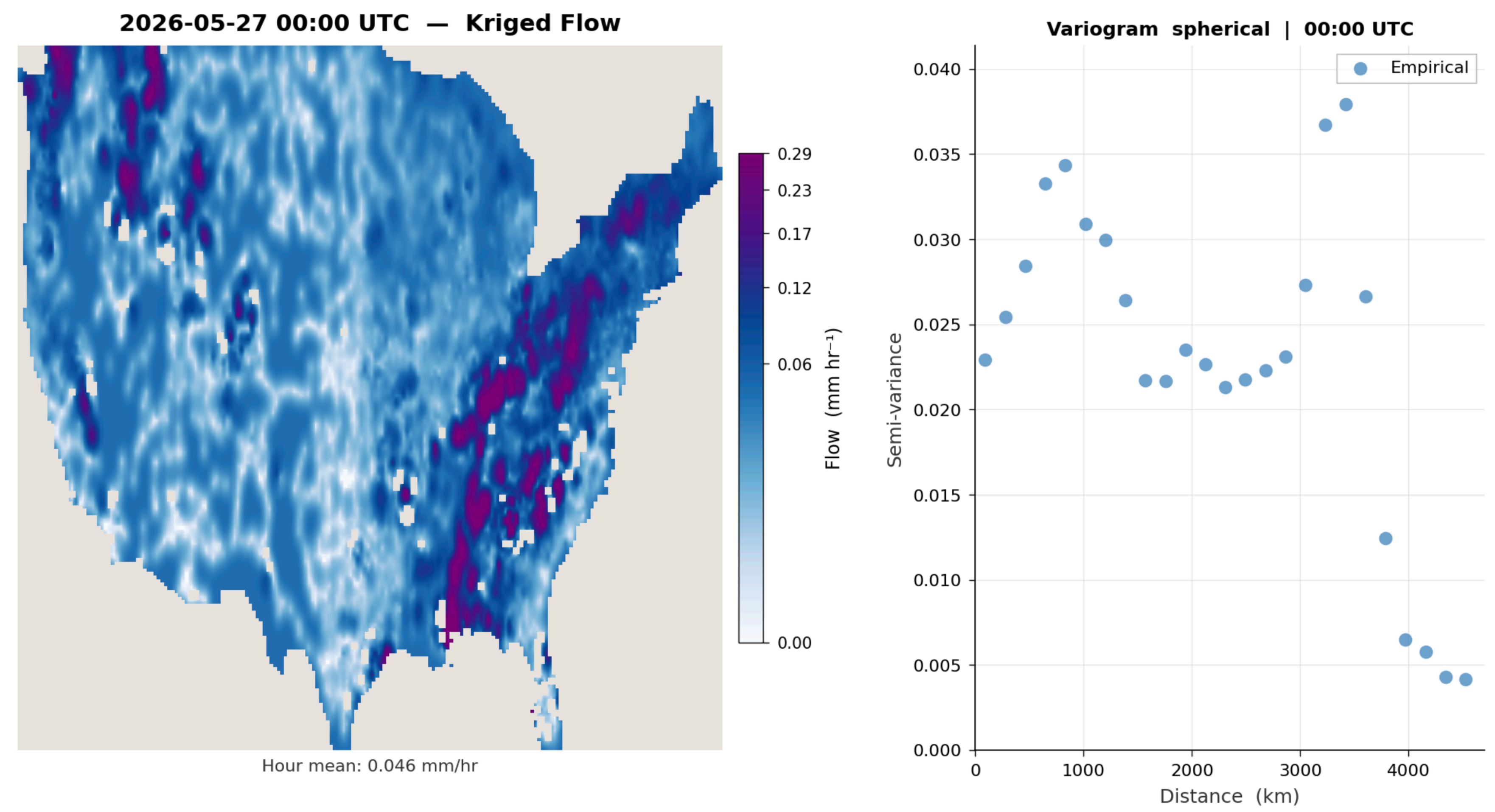

Figure 1 Empirical semivariogram (spherical model) for the 2026-05-27 00:00 UTC hour (mean = 0.046 mm/hr). Semivariance increases with distance (km) up to the sill, defining the correlation range used for kriging. Computed from gauge observations streamed from NRDS.

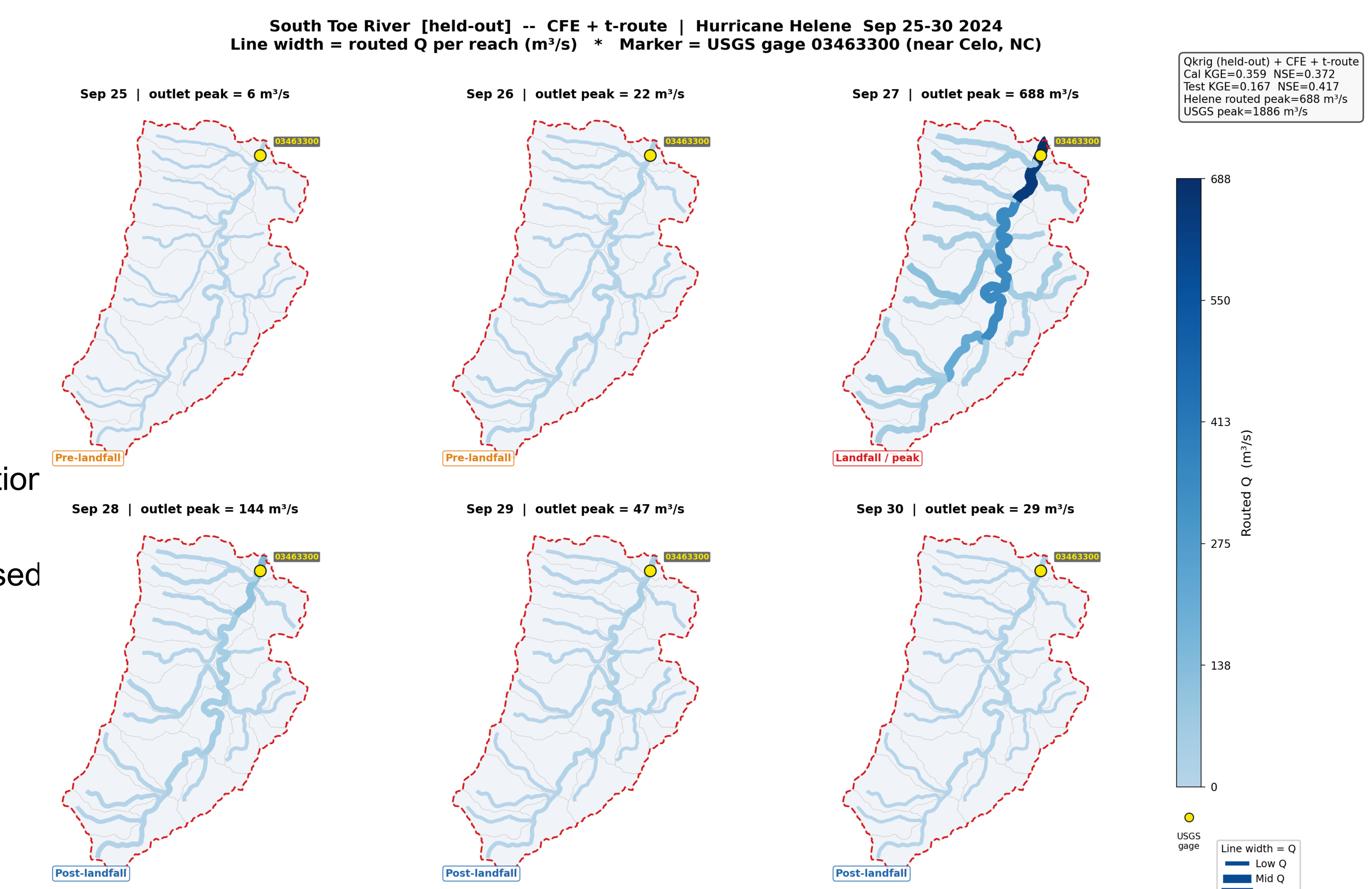

Figure 2 CFE calibrated on Qkrig precipitation + T-route Muskingum-Cunge routing · Period: Sep 25–30, 2024 Cal: KGE 0.359 / NSE 0.372 · Test: KGE 0.167 / NSE 0.413 Helene peak, modeled: 688 m³/s · observed (USGS): 1,886 m³/s

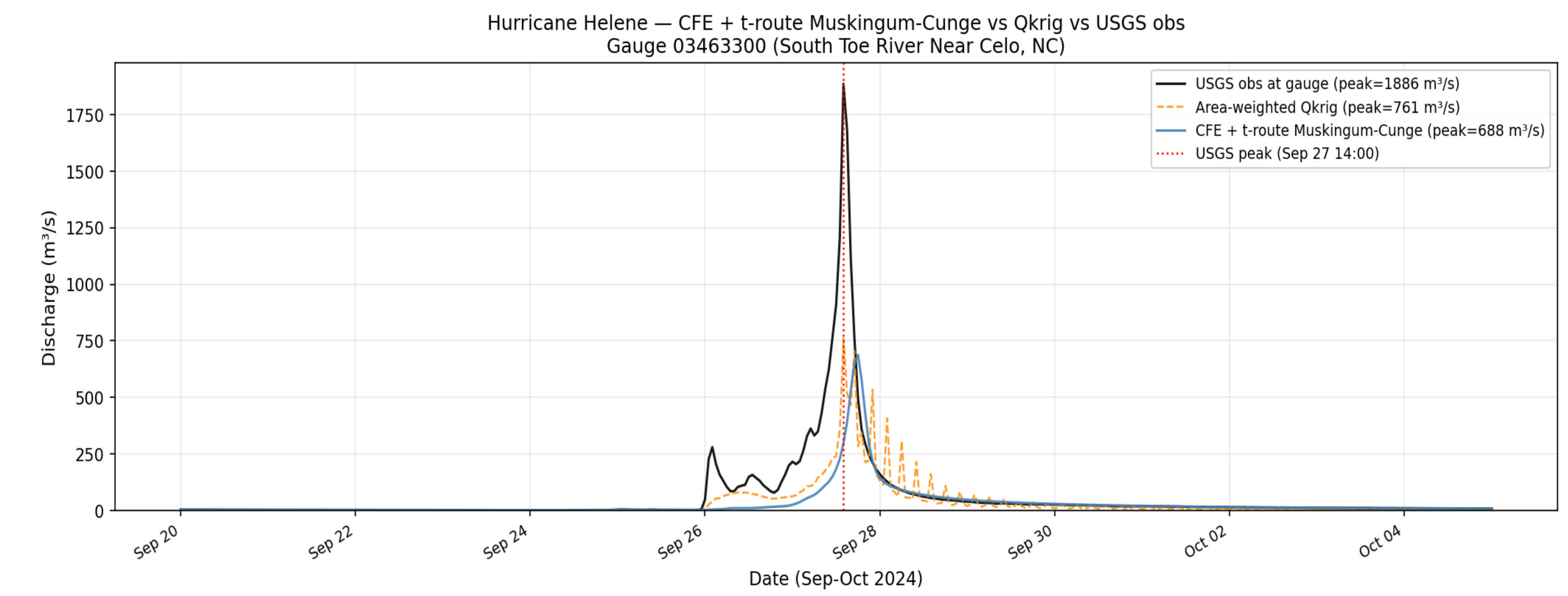

Figure 3 Hourly hydrograph at gauge 03463300 (South Toe R., Celo NC). Black: USGS obs (peak 1886 m³/s). Orange dashed: area-weighted Qkrig (peak 761 m³/s). Blue: CFE + t-route Muskingum-Cunge (peak 688 m³/s).

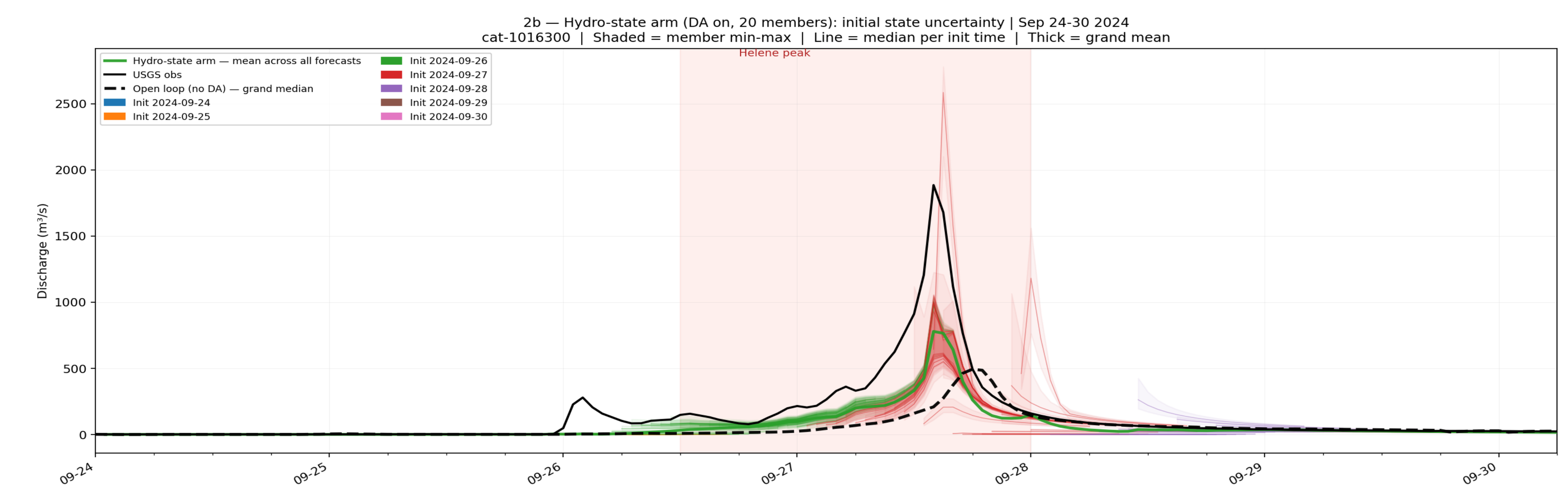

Figure 4 Green ensemble (DA on, 20 members): CFE with EnKF state corrections, colored by initialization date. Black dashed: open loop (no DA), grand median. DA-corrected states drive a clear ensemble response to Helene while the open loop captures limited flow. USGS observed peak: 1,886 m³/s.

Key Contributions

✓ Hourly kriging is OPERATIONAL on NRDS first-class datastream at CIROH-UA.

✓ Spatially complete coverage at 831,777 catchments × 24 hr daily.

✓ Pipeline scales to ~7,300 gauges × 831,777 catchments at ~12-min nightly runtime.

✓ CFE+ t-route ensemble members capture observed Helene peak (1886 m³/s) with quantified uncertainty.

✓ Kriging error variance provides observation-driven uncertainty at every estimation point.

Future Work

→ Evaluate ensemble forecasts during extreme events and improve peak prediction, dynamic variogram parameter calibration.

→ Extend DA benchmarking with EnKF assimilation of kriged pseudo-observations into CFE.

This research was supported by the Cooperative Institute for Research to Operations in Hydrology (CIROH) with funding under award NA22NWS4320003 from the NOAA Cooperative Institute Program. The statements, findings, conclusions, and recommendations are those of the author(s) and do not necessarily reflect the opinions of NOAA.

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